

Early Differentiation and Development



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Development of the Primary Embryonic Tissue Types after Fertilization

- * Cleavage
- * Blastulation
- * Gastrulation
- * Neurulation

Fertilization –

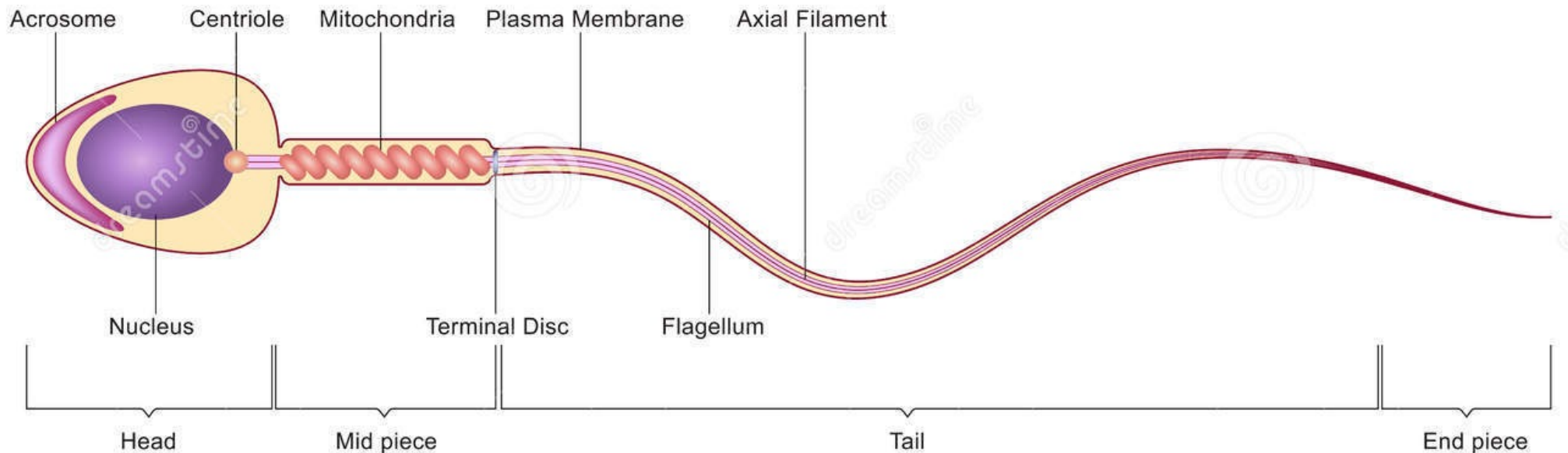
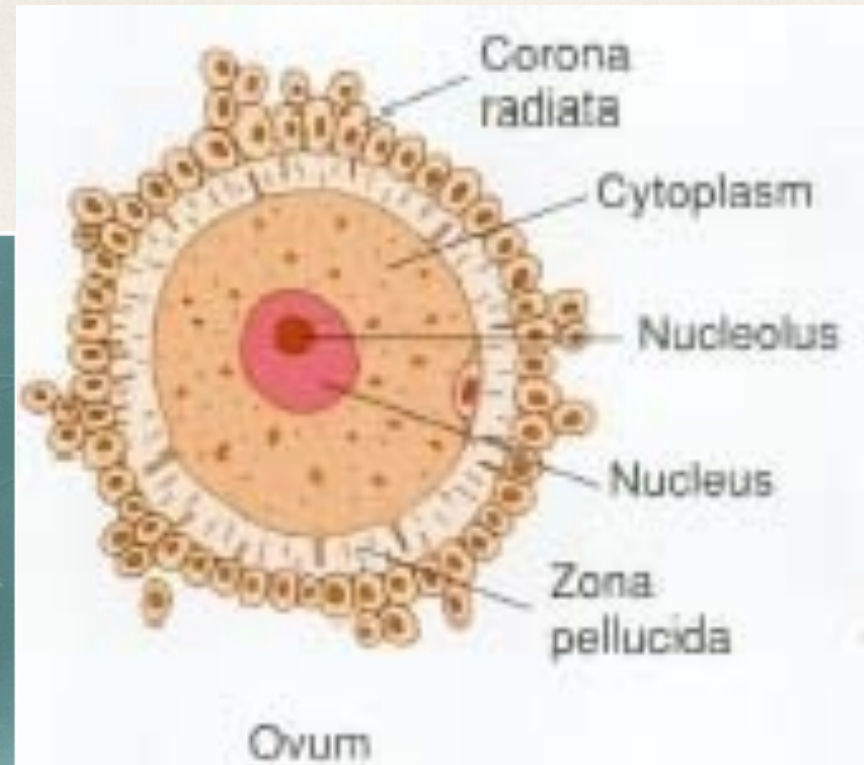
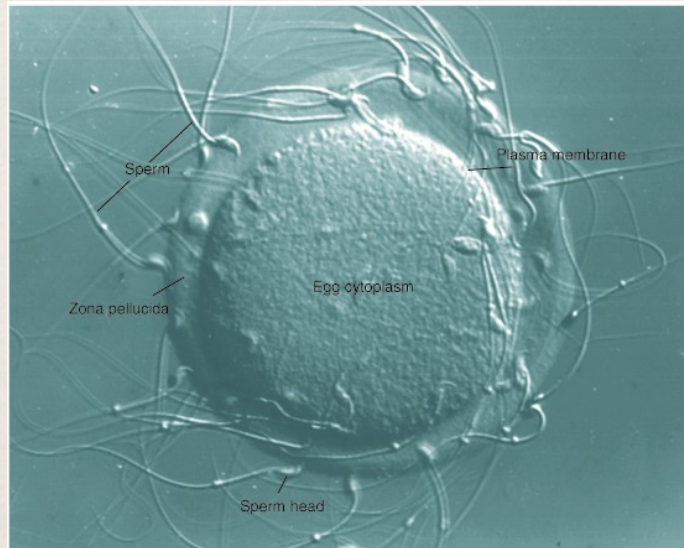
basic anatomy of the egg and sperm

Egg largest human
Cell

Sperm cell has 75-100
mitochondria

Egg has quarter of a
million

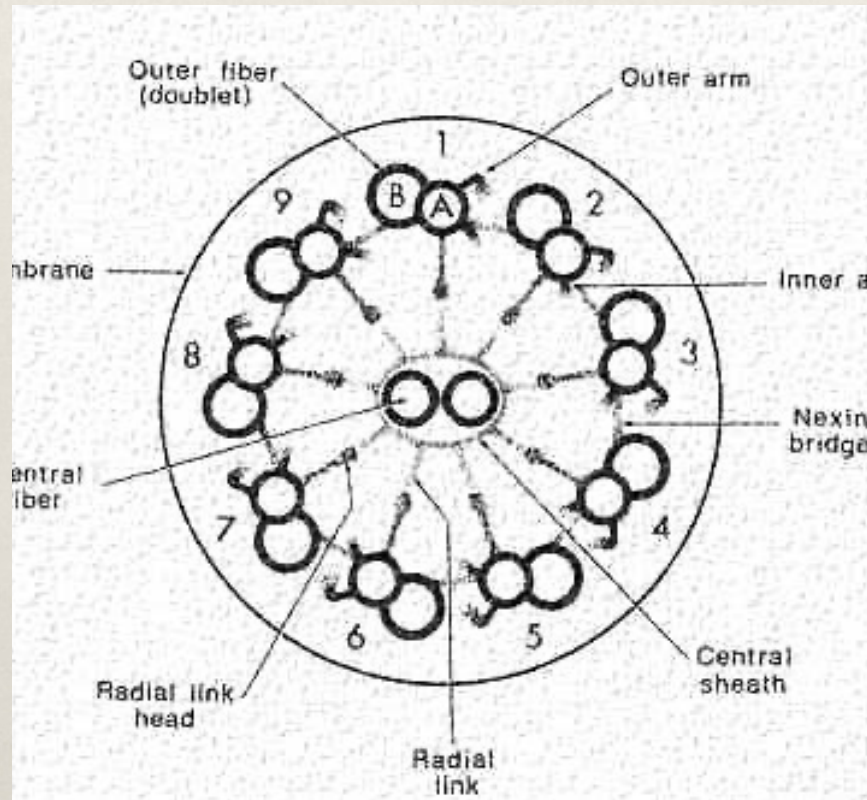
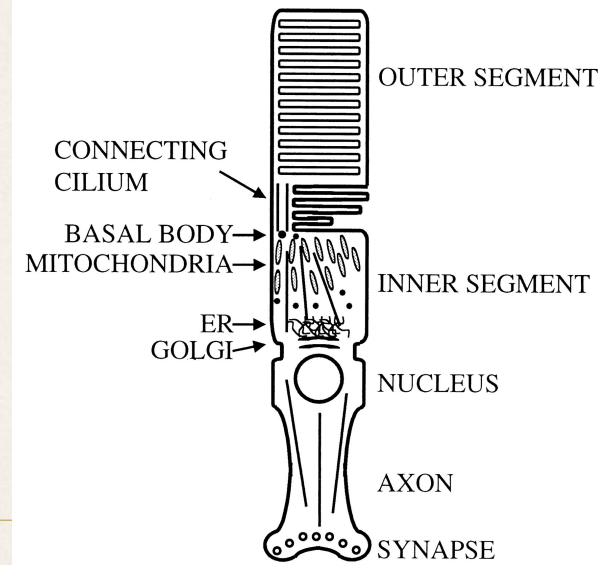
Mitochondrial DNA
(and therefore,
mitochondrial diseases)
are inherited from the
mother.



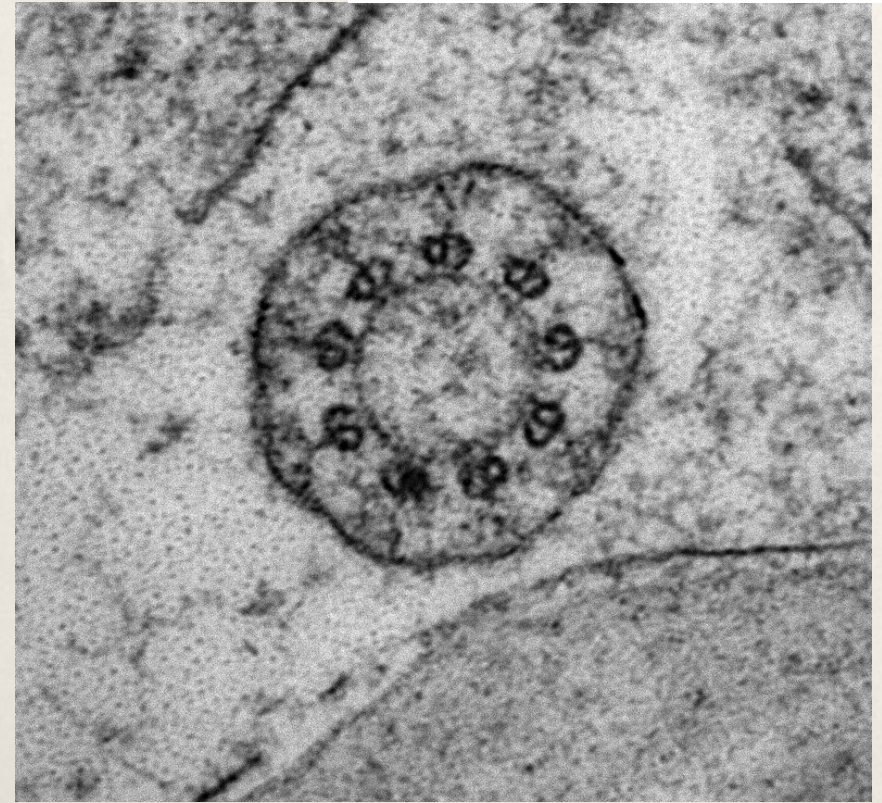
A note on microtubules within motile and non-motile Cilia

Motile cilia have a ring of 9 doublets and a central pair

Non-motile cilia have a ring of 9 doublets with no central pair



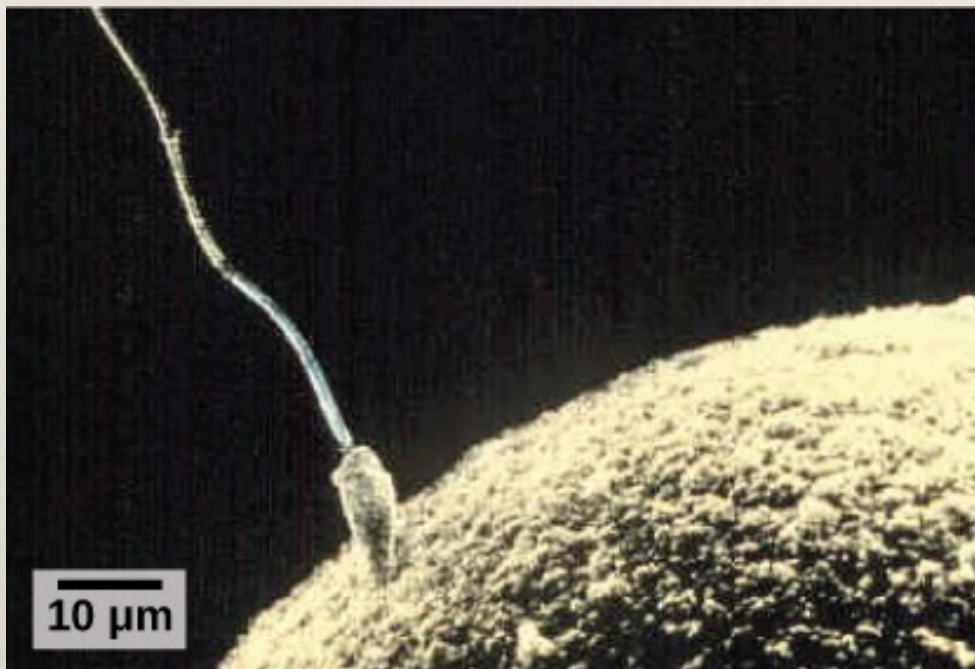
9 + 2 Sperm tail



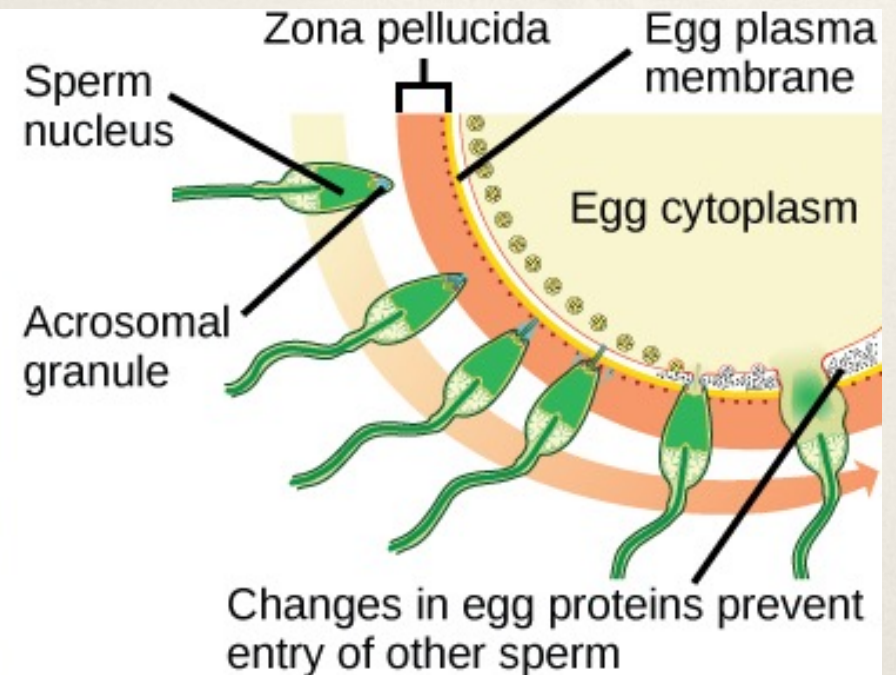
9 + 0 Retinal photoreceptor

Fertilization — 4 basic steps

- ★ Sperm binding in zona pellucida
- ★ Acrosome reaction – release of acrosomal enzymes into zona pellucida, digesting it and allowing sperm plasma membrane to reach egg plasma membrane.
- ★ Cortical reaction – sperm binding to plasma membrane causes release of cytoplasmic granules bearing enzymes. These enzymes, when released into zona pellucida specifically digest the portion of the zona pellucida glycoprotein that binds sperm. This prevents additional sperm from binding to prevent poly-spermy.
- ★ Delivery of male chromatids into the egg.



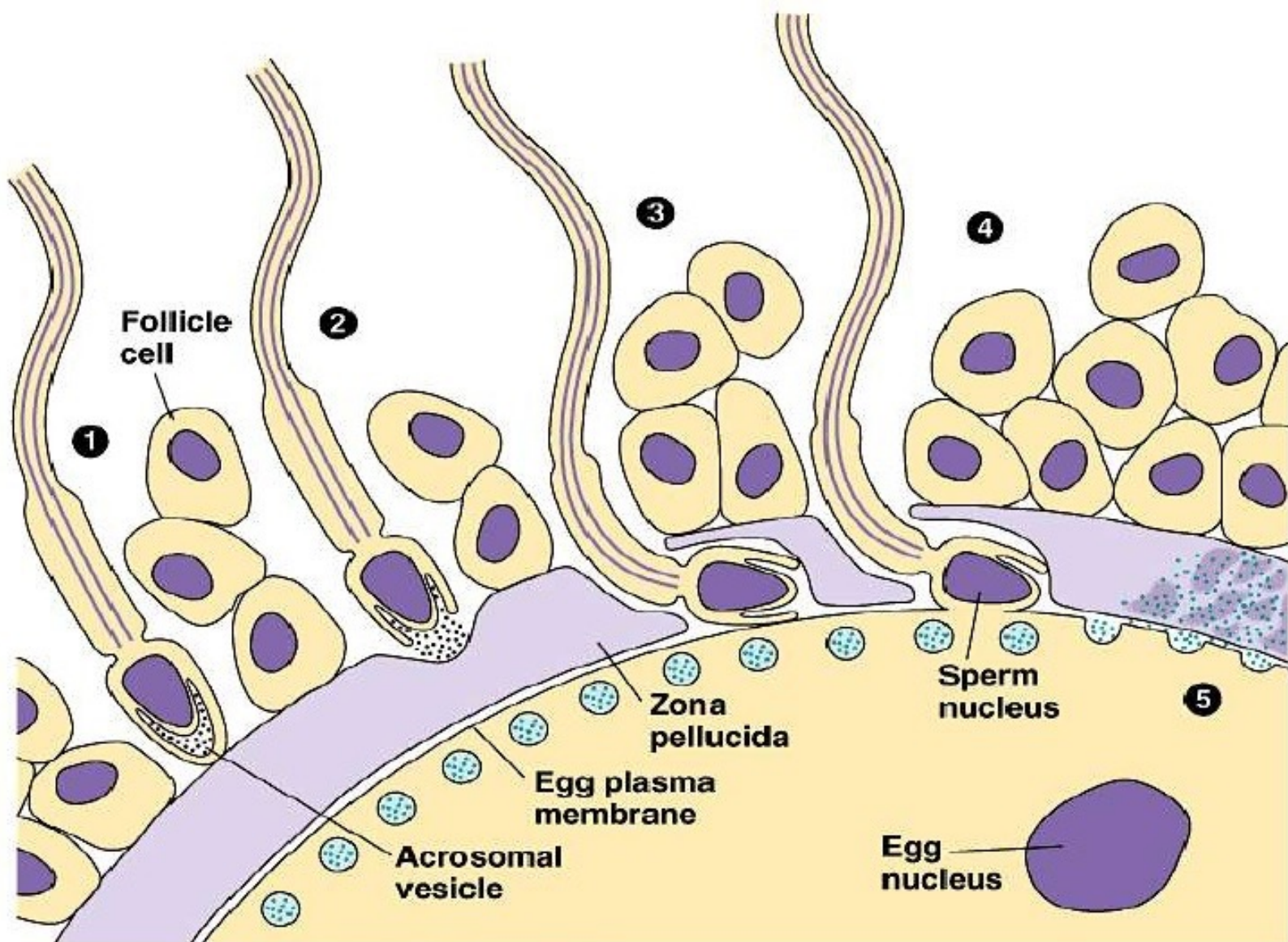
(a)



(b)

Fertilization

Fusion of the egg and sperm membranes



Cleavage

After fertilization, the DNA of the egg and the sperm merge into a ZYGOTE, which remains enveloped within the zona pellucida. During this period of rapid division, no time is spent on growth. Hence the overall size of the zygote remains the same.

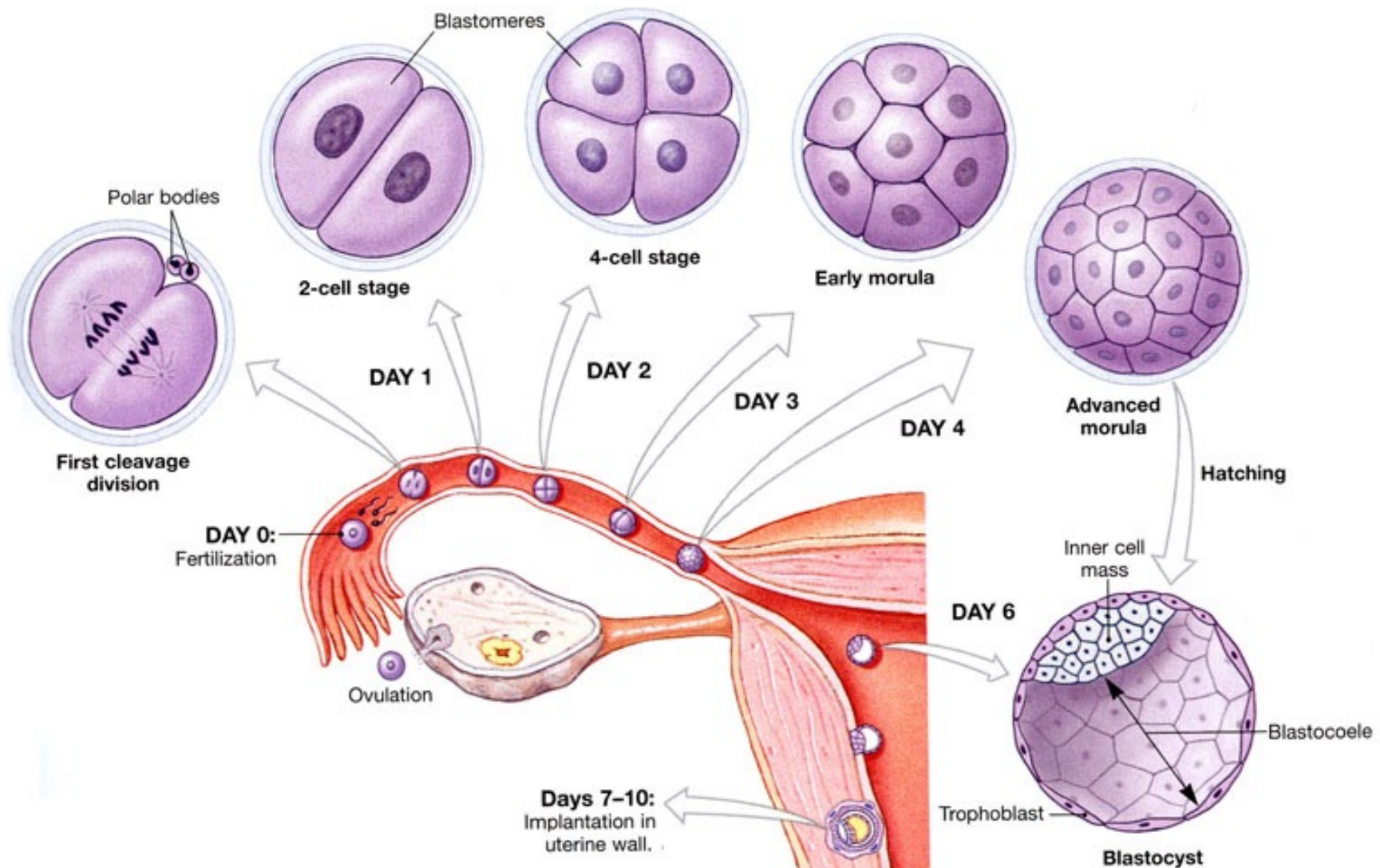
1-2, 2-4, 4-8, 8-16, 16-32 in rapid succession. At the 32 cell stage (lower middle color image) the name changes from zygote to MORULA. Morula formation defines the end of the CLEAVAGE stage. As division continues, the cells undergo compaction (lower right).



MORULA is
LATIN for
Mulberry

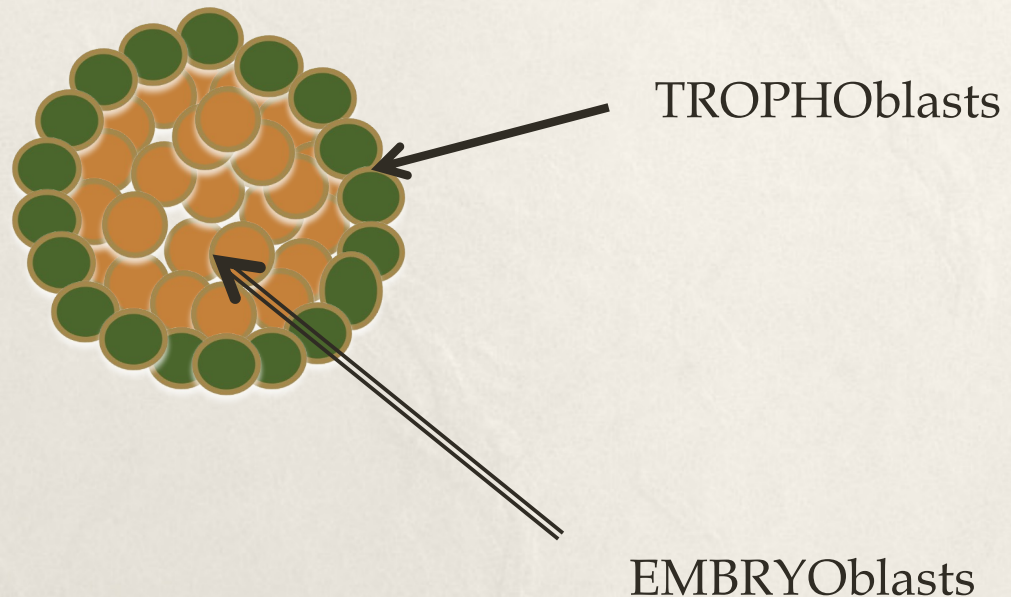


The cleavage stage occurs during passage along the Fallopian tube



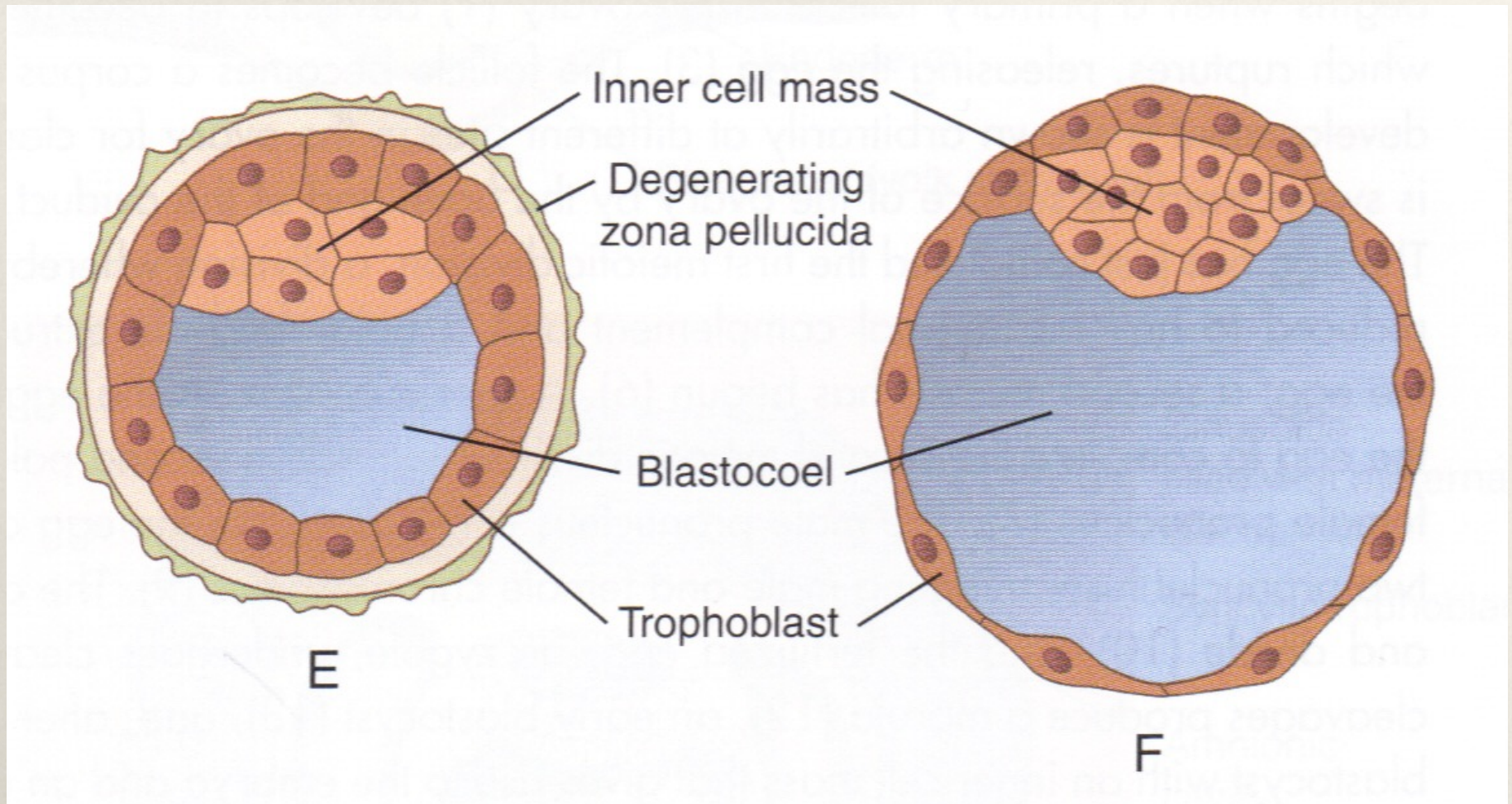
BLASTULATION

- * As COMPACTION continues, the first differentiation from a uniform cell mass occurs. A single layer of cells, different from the “inner cell mass”, comes to surround the compacted morula.

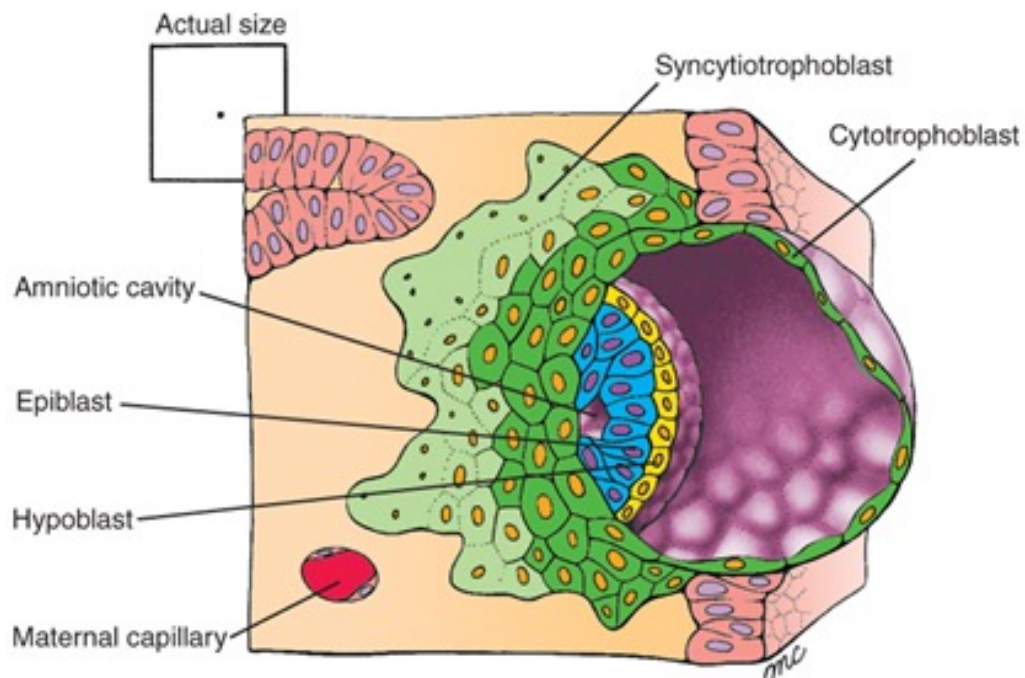


BLASTULATION

As further compaction of the inner cell mass, within the sphere of trophoblasts occurs, the inner mass of embryoblasts shifts to one pole. In this process, a cavity called the blastocoel develops. The stage shown in F below is the BLASTOCYST. It is at this stage, after the zona pellucida has dissolved, that implantation usually occurs.

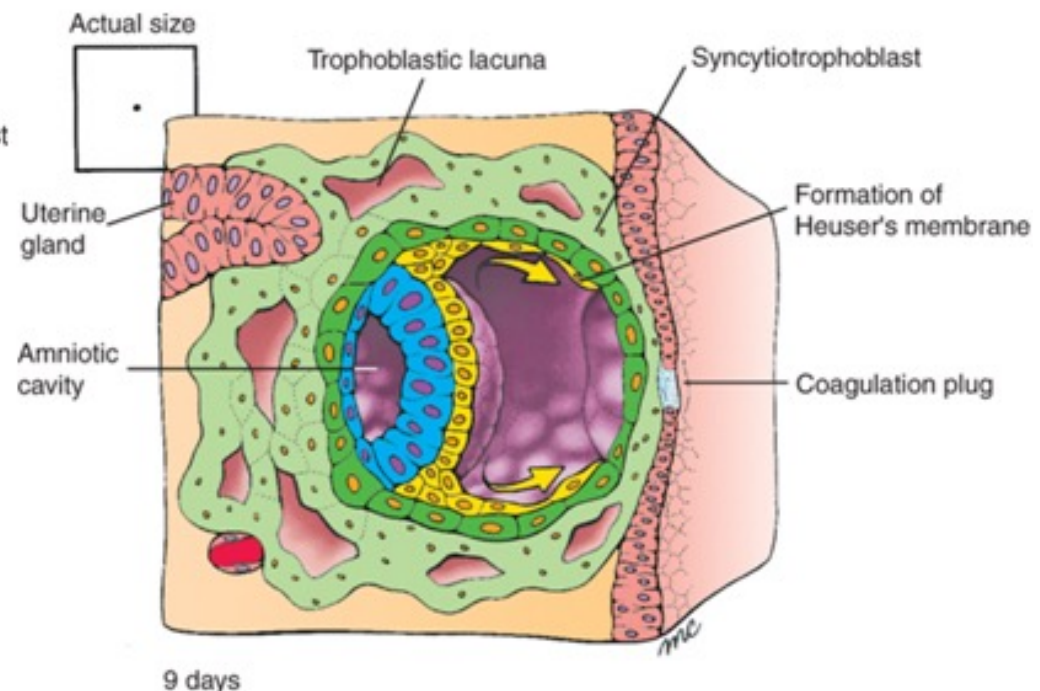


As implantation occurs (about day 8-9), the inner cell mass of embryoblasts hollows out another cavity, the amniotic cavity. The row of cells closest to the blastocyst cavity differentiate into a single plate of cells termed the **HYPOBLASTS**. The remaining cells of the inner cell mass, that are cleaving out the amniotic cavity are now termed the **EPIBLASTS**



8 days

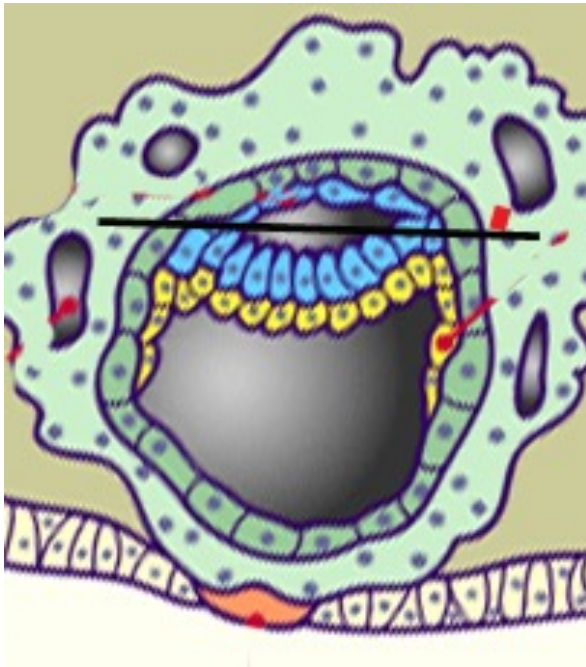
Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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9 days

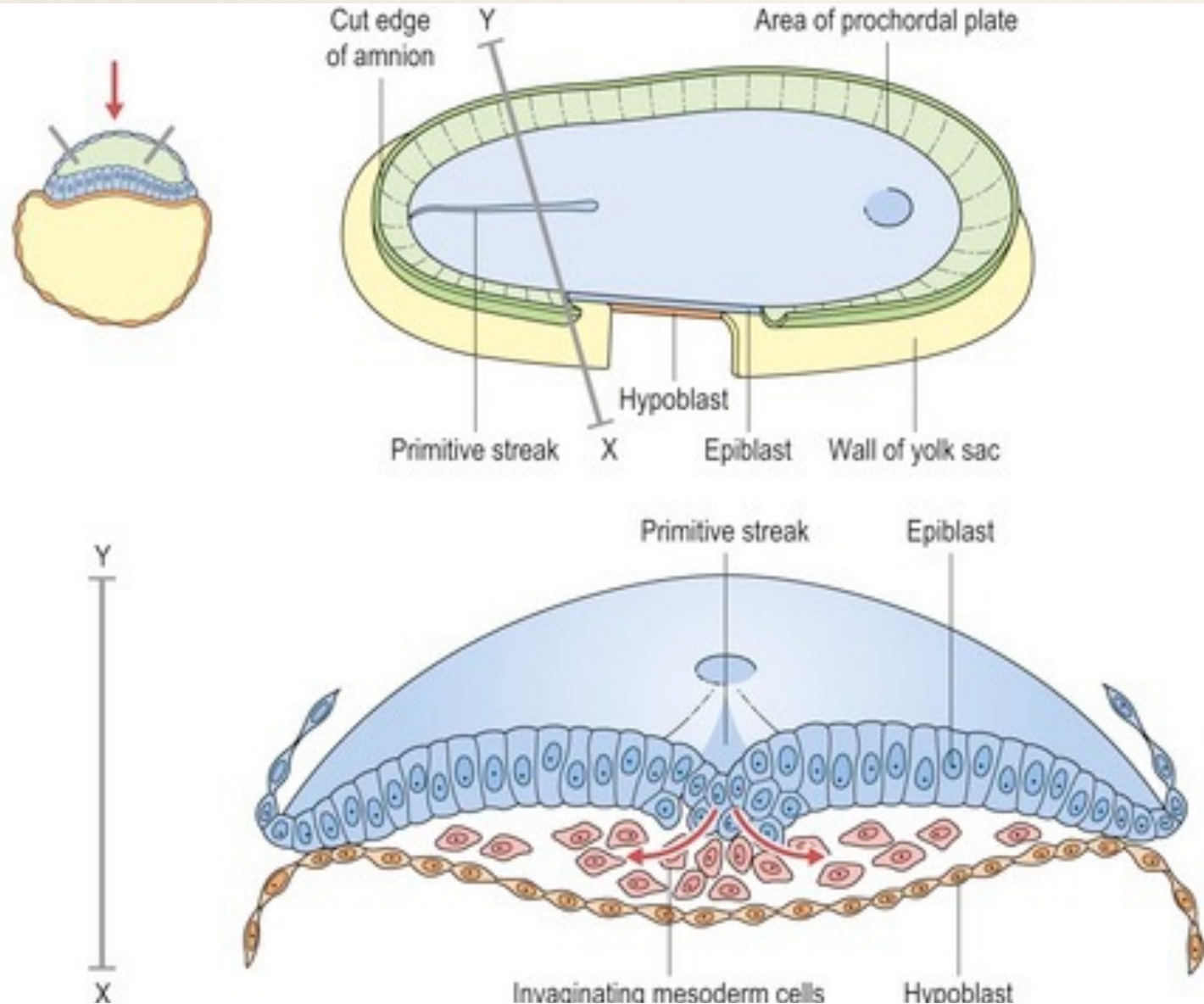
Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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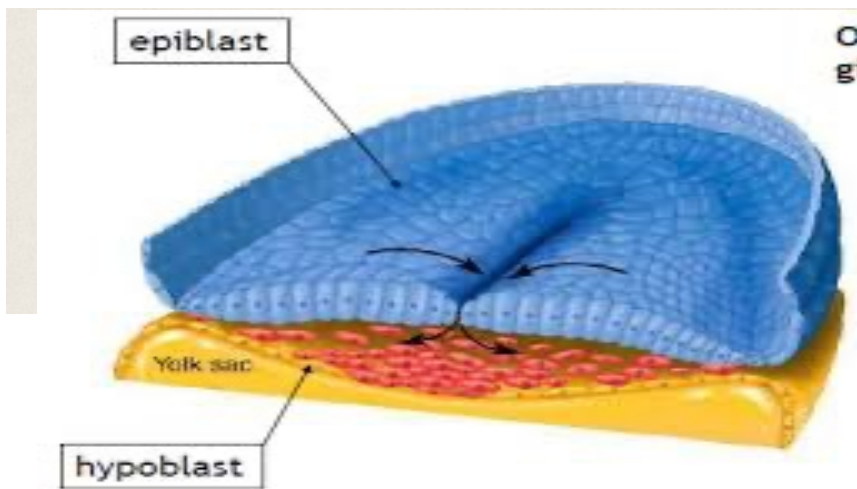
The Bilaminar Disc



We have compressed down to two layers of cells separating the blastocoel from the amniotic cavity. An “upper” layer of epiblast cells and a lower layer of hypoblast cells.

A linear, focal proliferation occurs along the epiblastic layer called the **PRIMITIVE STREAK**. This line establishes top, bottom, left and right.





GASTRULATION

Ends with a TRILAMINAR disc

GERM LAYER

Ectoderm

Endoderm

Mesoderm

BECOMES

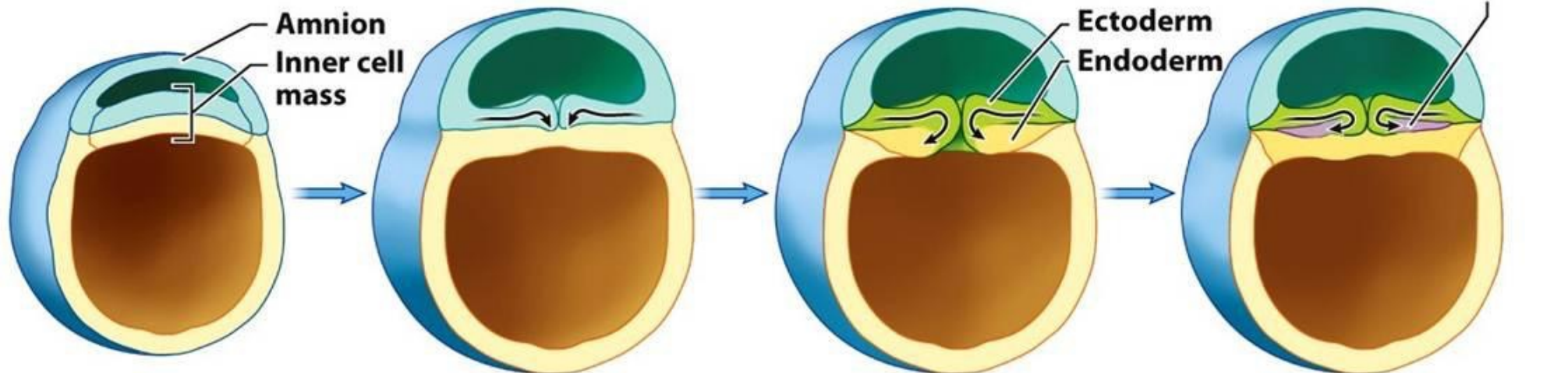
Outer layer of skin, hair, lining of the nose and mouth, and the nervous system

Digestive tract, respiratory tract, liver, and pancreas

Muscles and skeleton

GASTRULATION

The second phase of development is gastrulation, a migration of blastocyst cells inward, leading to multiple distinct layers of tissue called germ layers.



Cells of the inner cell mass begin to differentiate as the amnion forms

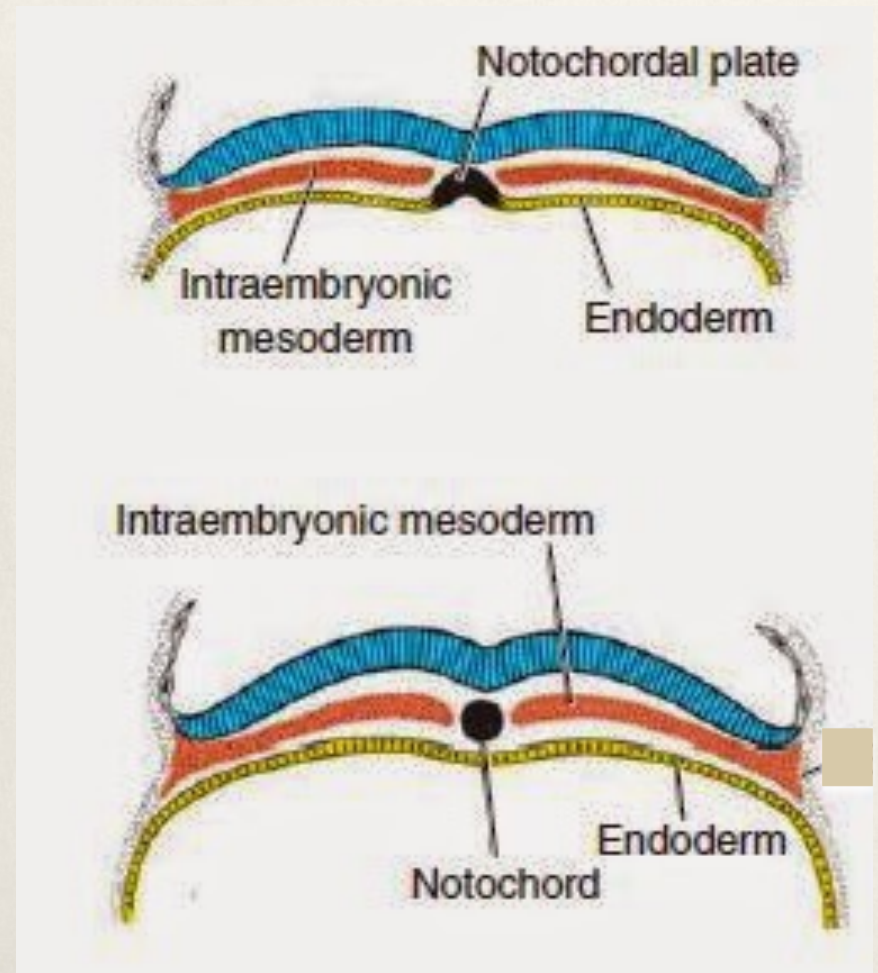
A gastrula develops when cells begin to migrate inward, forming an indentation.

The cells continue to push inward, forming the endoderm. Cells that remain on the outer surface of the gastrula are called ectoderm.

The mesoderm is formed, as additional cells migrate inward between the endoderm and ectoderm.

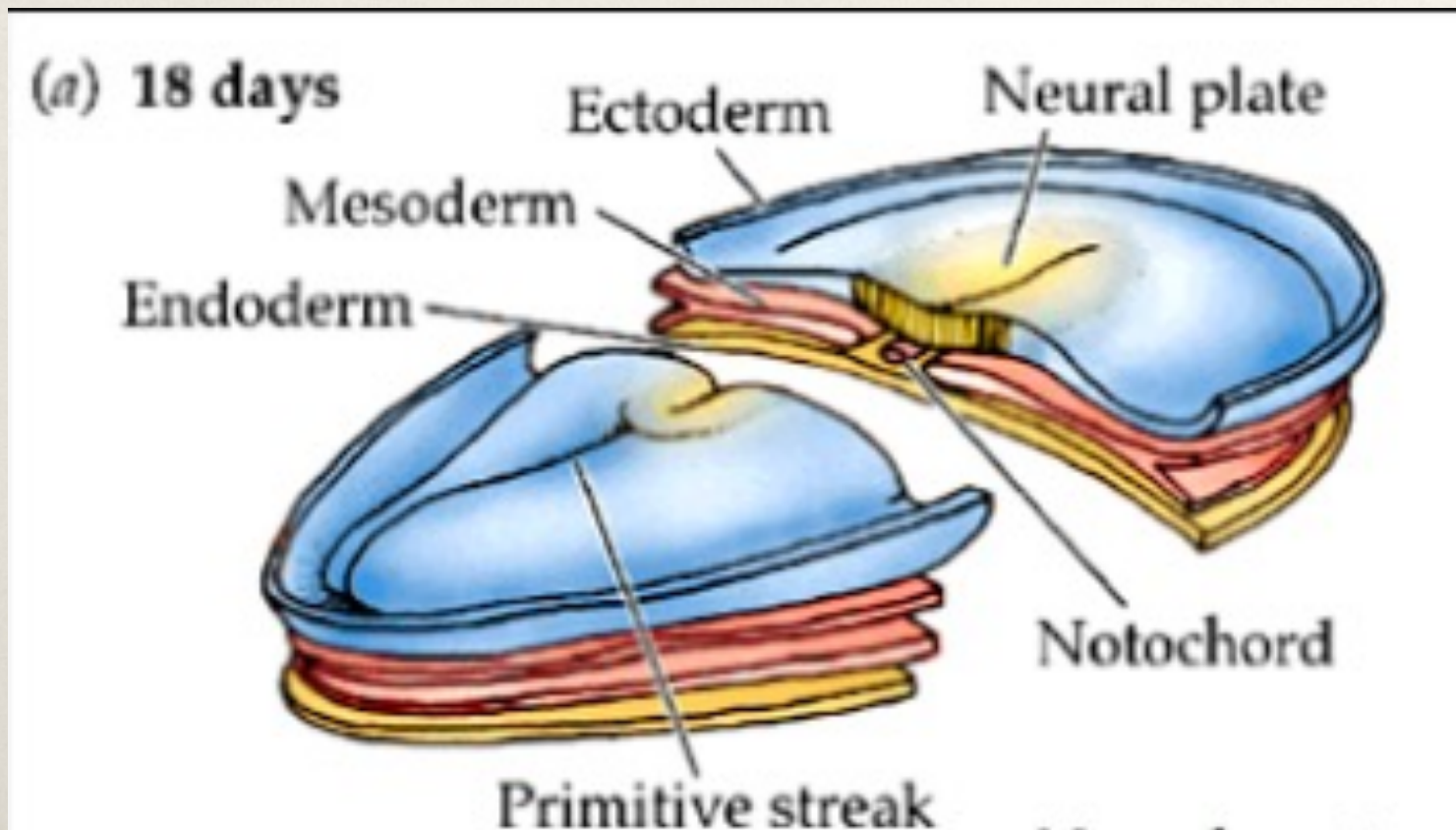
NEURULATION – Notochord and Neuroectoderm formation

- * In the mesoderm underlying the primitive streak, a chord of tissue develops called the notochord. The notochord establishes the plane along which the spinal cord and vertebral column will form. These do NOT form by building on the notochord but by using it as an orientation marker. The notochord contributes only to the intervertebral discs in the adult.



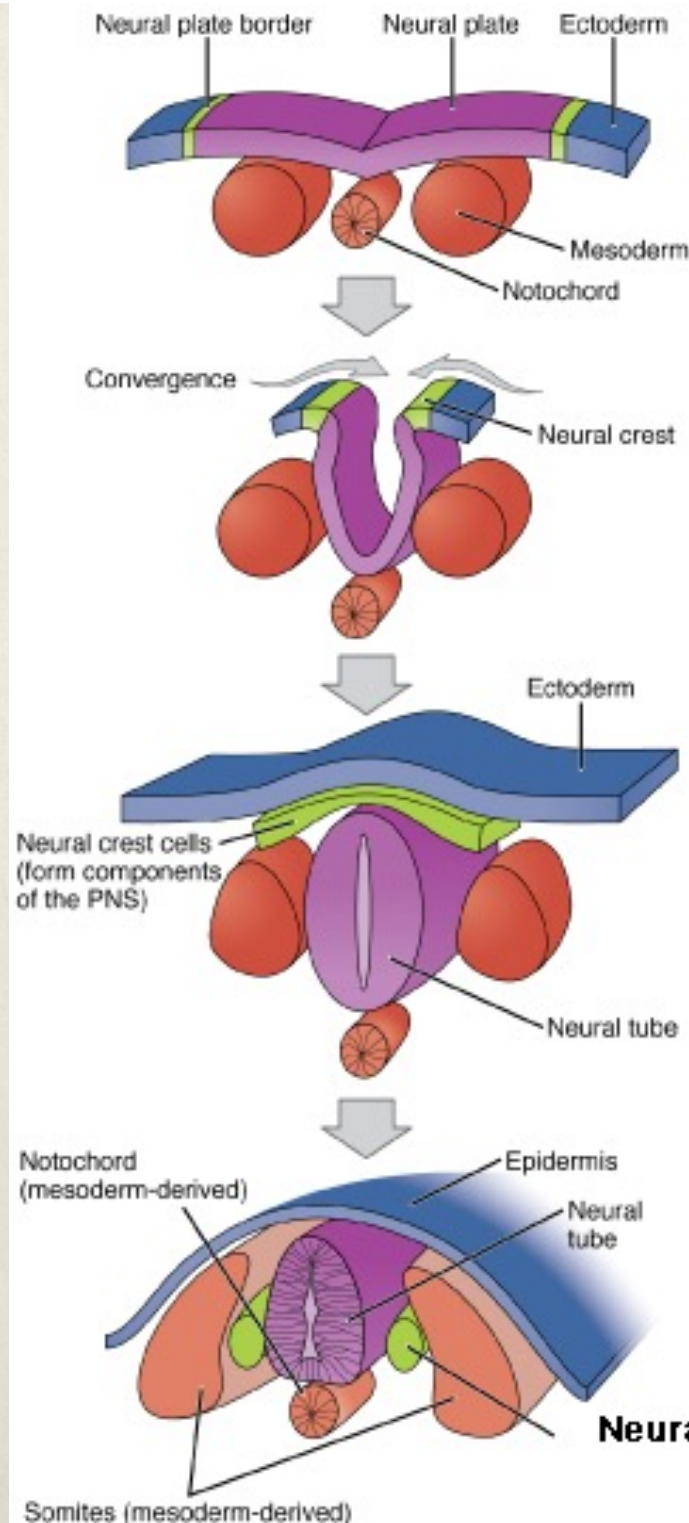
NEURULATION – Notochord and Neuroectoderm formation

- * The presence of the notocord induces differentiation and proliferation of a linear strip of surface ectoderm overlying it in the midline.
- * The thickened plate of surface ectodermal cells is called the neural plate and the portion of surface ectoderm forming this plate is now called NEUROECTODERM



Formation of Neural Groove, neural crest and closure of the neural tube

- ★ Important to note that neural groove closes first at its midpoint and then closes up and down from that point



- ① Neuroectodermal tissues differentiate from the ectoderm and thicken into the neural plate. The neural plate border separates the ectoderm from the neural plate.
- ② The neural plate bends dorsally, with the two ends eventually joining at the neural plate borders, which are now referred to as the neural crest.
- ③ The closure of the neural tube disconnects the neural crest from the epidermis. Neural crest cells differentiate to form most of the peripheral nervous system.
- ④ The notochord degenerates and only persists as the nucleus pulposus of the intervertebral discs. Other mesoderm cells differentiate into the somites, the precursors of the axial skeleton and skeletal muscle.

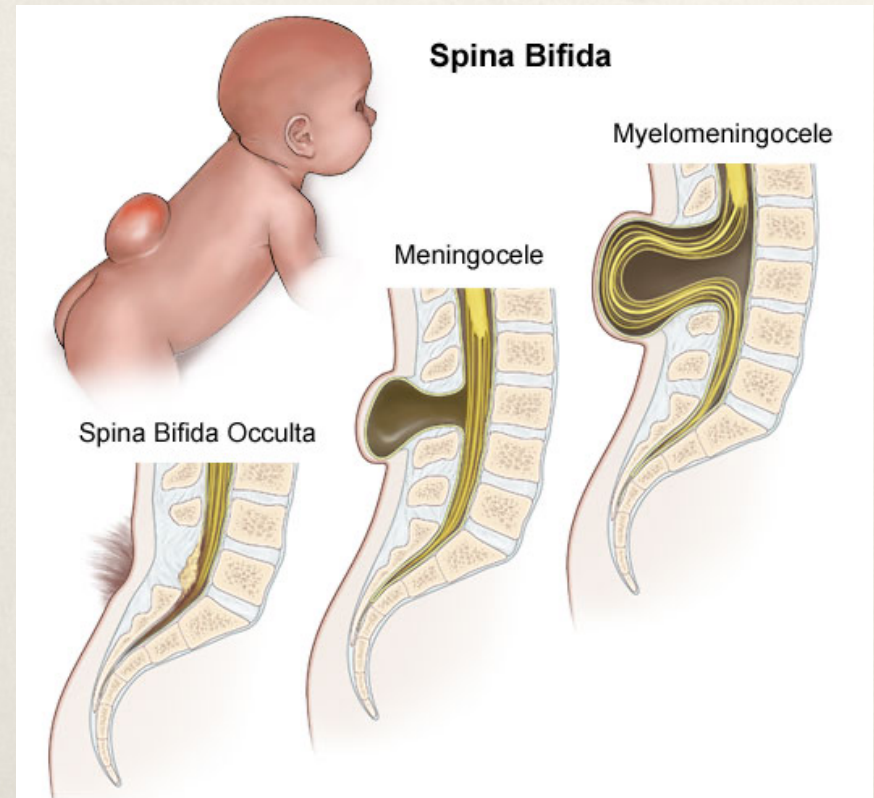
Failure of Neural Tube Closure

If the neural tube fails to completely close above the mid-point – **anaencephaly** (absence of a major portion of the brain skull and scalp)



Generally
incompatible
with life

If the neural tube fails to completely close below the mid-point – **spina bifida**
Meningocele – a bulge in the dura containing no spinal cord tissue
Myelomeningocele – a bulge in the dura containing spinal cord tissue



Compatible with life, with varying
degrees of physical disability

Analogy in the Eye

ECTASIA

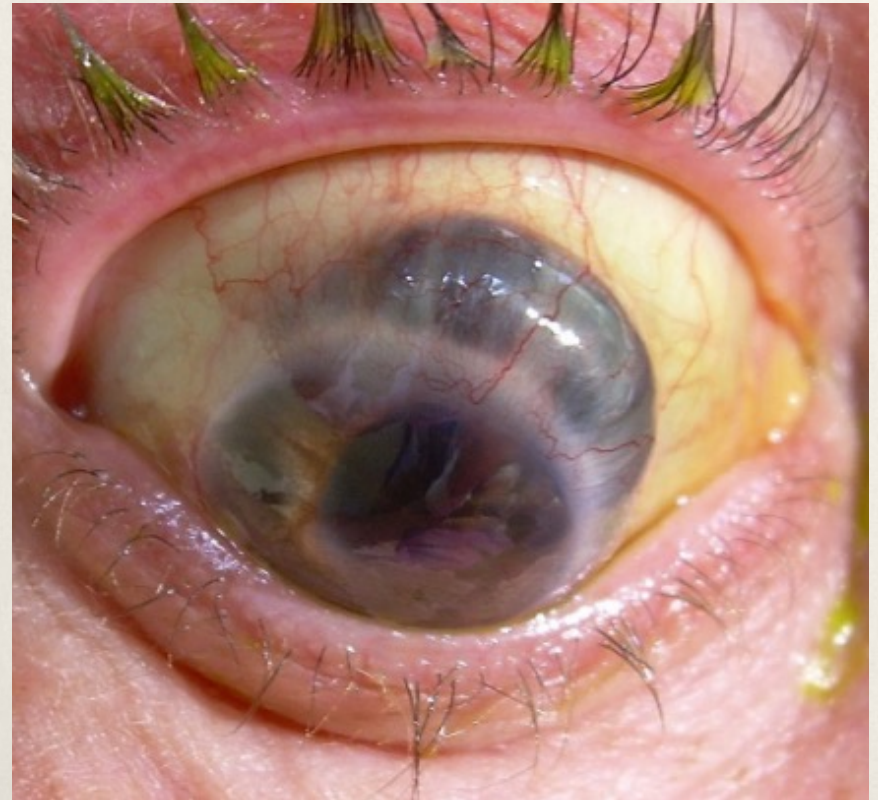
A bulge of the eye wall that does NOT contain uveal tissue



Thinning and bulging of the cornea from keratoconus, an ocular ectasia

STAPHYLOMA

A bulge of the eye wall that DOES contain uveal tissue



Thinning and bulging of the sclera containing uveal tissue, an ocular staphyloma

NEURULATION - Summary



NEURULATION

GERM LAYER

 Ectoderm

 Endoderm

 Mesoderm

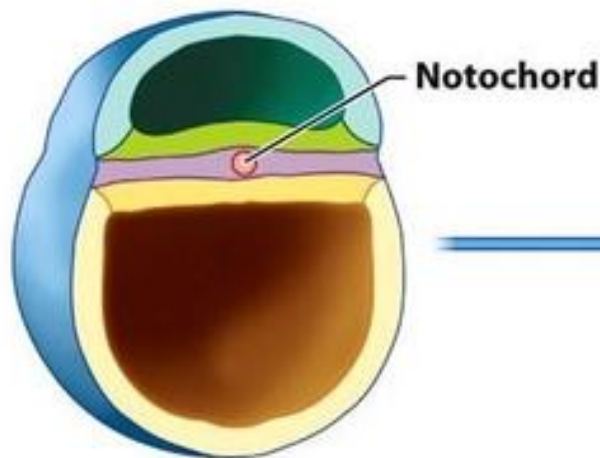
BECOMES

Outer layer of skin, hair, lining of the nose and mouth, and the nervous system

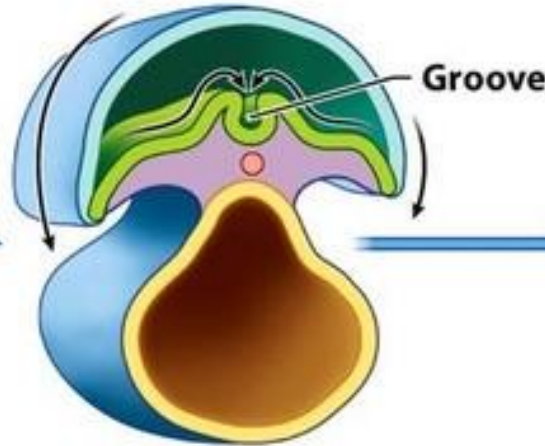
Digestive tract, respiratory tract, liver, and pancreas

Muscles and skeleton

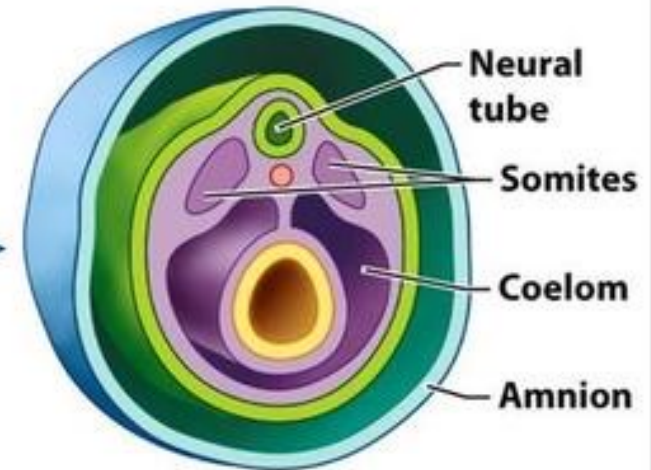
During the third week of development, the three types of tissues formed during gastrulation begin to develop into the various organs and tissues of the organism.



Cells within the mesoderm form the notochord, which runs the length of the embryo.



The ectoderm folds inward, forming a groove that runs the entire length of the embryo.



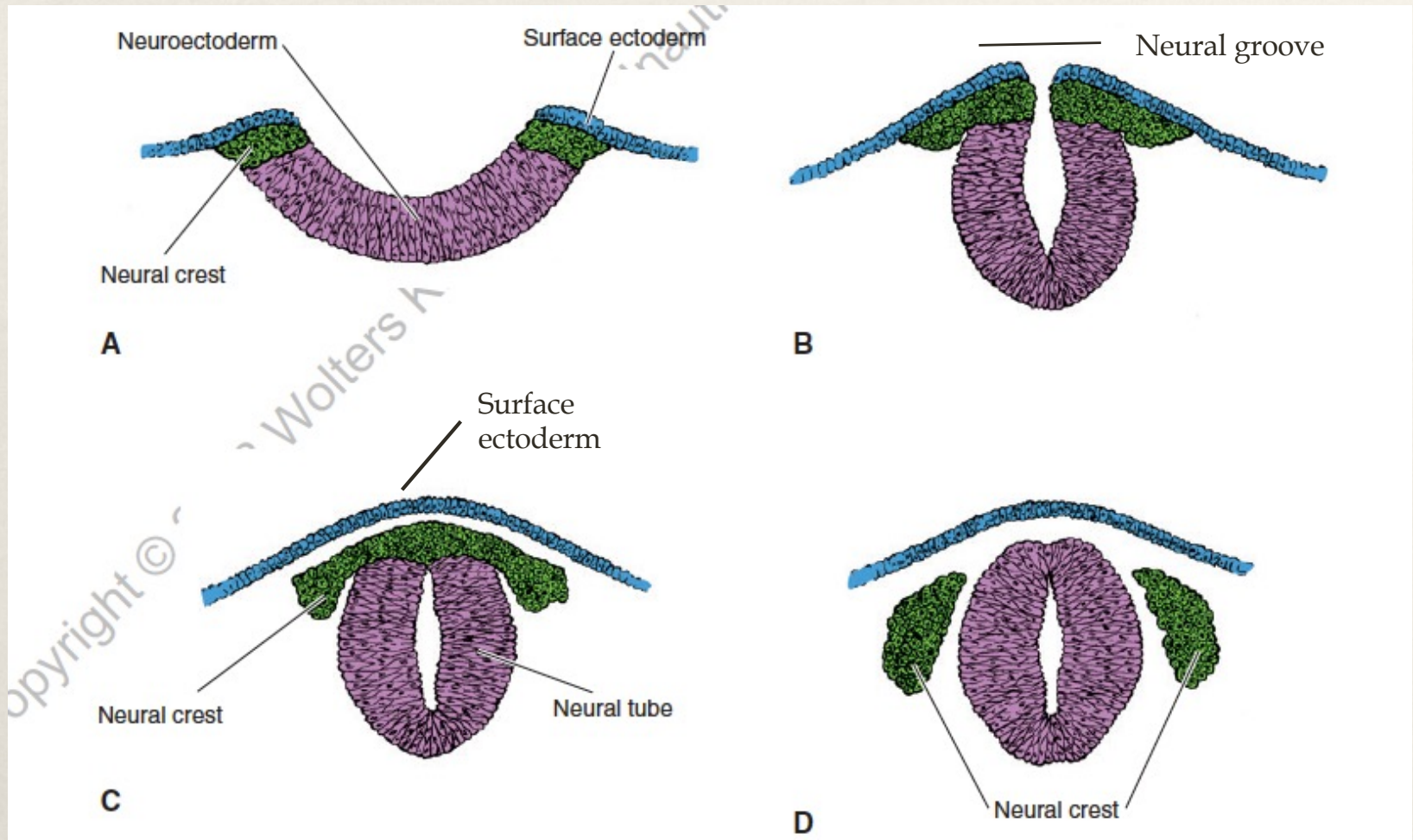
Neurulation is completed with the formation of the neural tube. Somites and the coelom develop from the mesoderm.

Figure 25-22

What Is Life? A Guide To Biology

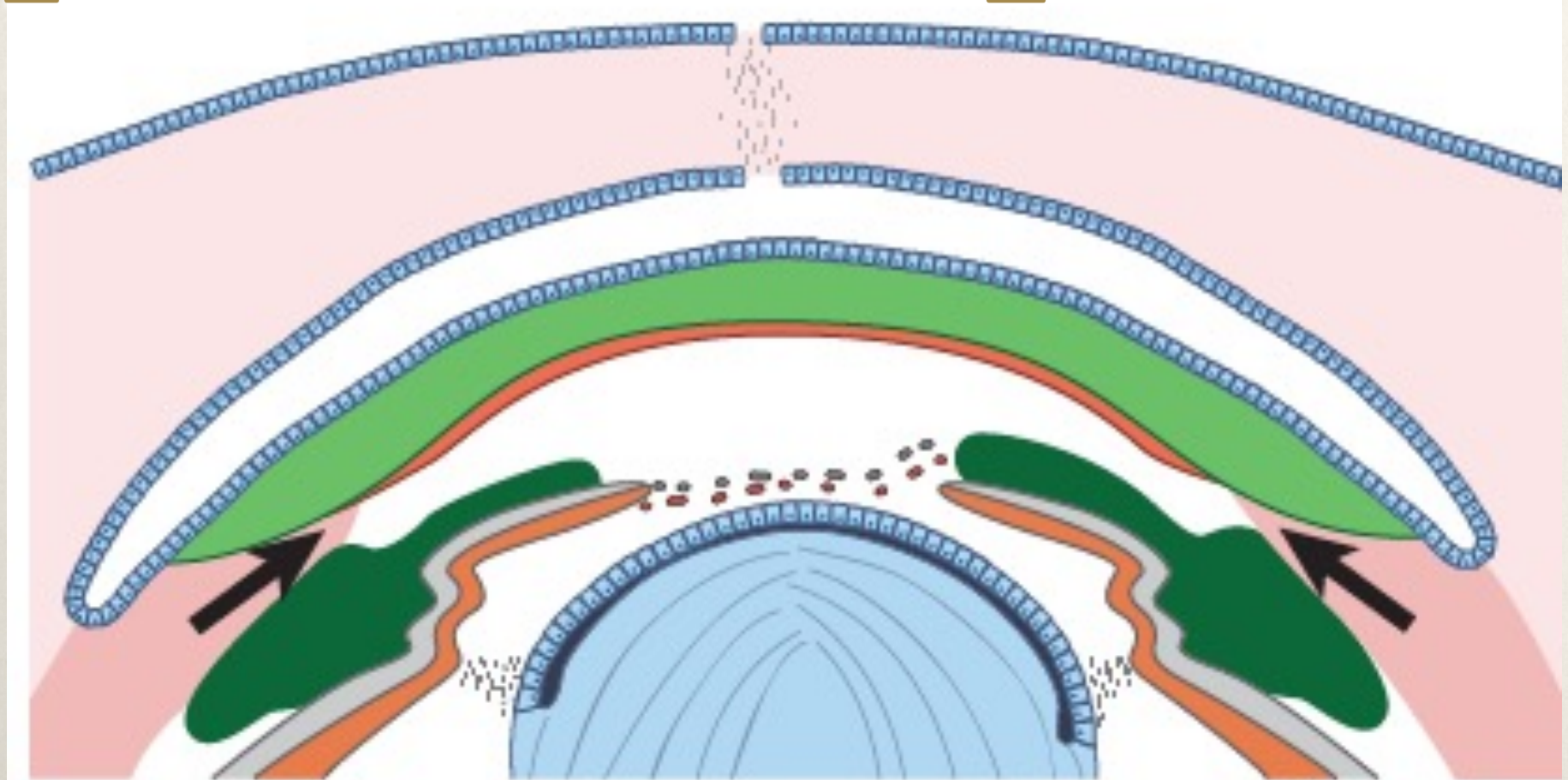
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Formation of Neural Crest Tissue



The Eye – contributions from surface ectoderm, neuroectoderm, mesoderm and neural crest. But NOT endoderm.

	Surface ectoderm		Neural crest first wave
	Inner neuroectoderm		Mesoderm
	Outer neuroectoderm		Neural crest 2 nd wave - ant
			Neural crest 2 nd wave - post

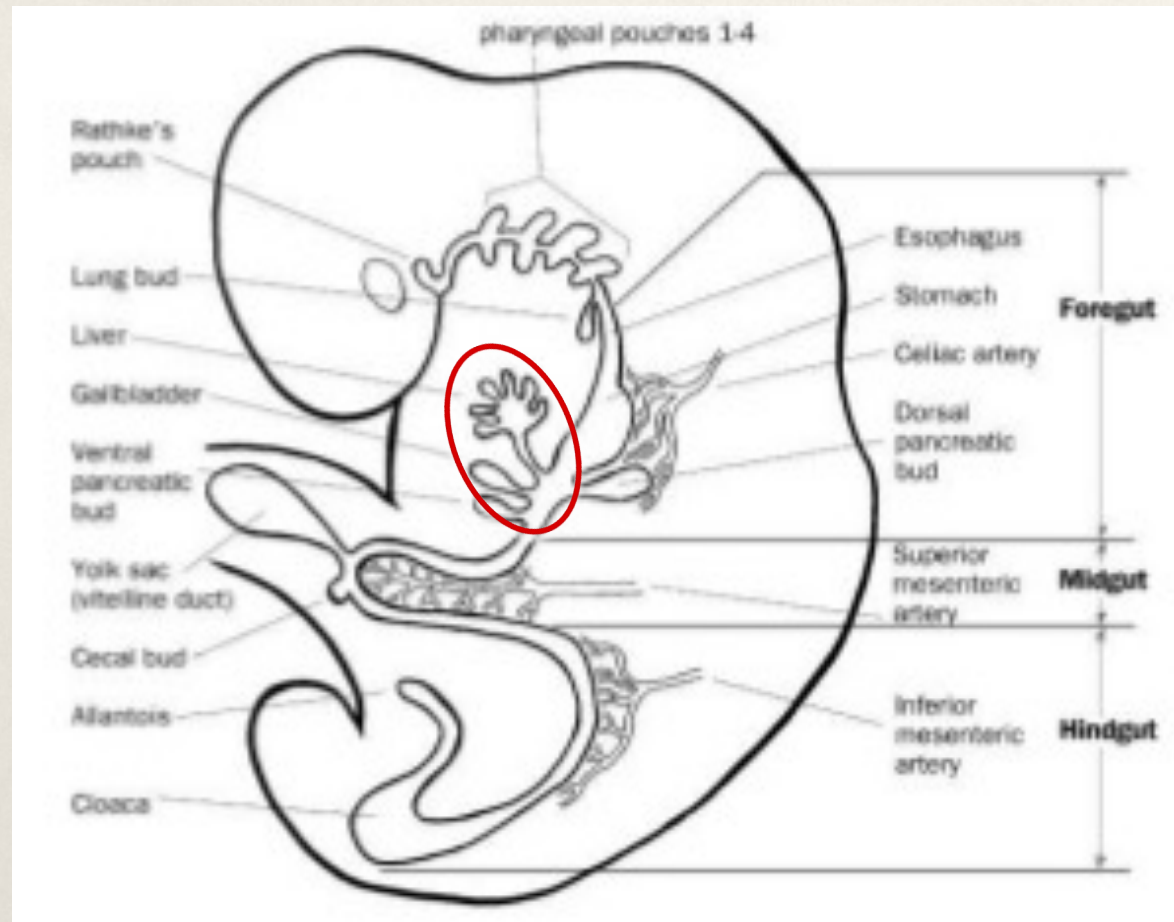


Overview

- * Endoderm – Upper respiratory tract including lungs and GI tract and associated exocrine glands including liver and pancreas
- * Mesoderm – muscles including heart, skeletal system and genital and urinary tracts
- * Surface ectoderm – skin and adnexal structures
- * Neuroectoderm - CNS

Exocrine Glands – from liver to Meibomian glands

- * Development of exocrine glands begins at the epithelial (ectoderm or endoderm) surface to which its product will be delivered. This ensures that you do not form a gland and only then, try to find the target surface.
- * Whether liver or Meibomian gland, a rod of cells from the surface proliferates and drives into surrounding tissue. This rod branches and differentiates the glandular elements.
- * Finally the solid rod is hollowed out, connecting the product of this exocrine gland with its target surface, whether the lid margin or the lumen of the intestine via the bile duct.



Khan Academy Videos

- ★ Oogenesis

<https://www.khanacademy.org/science/health-and-medicine/human-anatomy-and-physiology/reproductive-system-introduction/v/spermatogenesis>

- ★ Spermatogenesis <https://www.youtube.com/watch?v=sGLezrEOeaE>

- ★ From Zygote to Neurulation

<https://www.khanacademy.org/test-prep/mcat/cells/embryology/v/early-embryogenesis-cleavage-blastulation-gastrulation-and-neurulation>